Momentum transport in the Gram metric of the mimetic projection: designs, identities and measurements

audit note, TwoPhaseAMR.jl

Abstract

The mimetic projection (`MimeticProjection`, metric MMd) measures the phase velocities in the Gram metric $\mathcal{M}$; the momentum transport still lives in the lumped metric W of the staggered phase masses. This note asks for a transport that is consistent with $\mathcal{M}$: exact $\mathcal{M}$-momentum, exact free-stream preservation with the volume-of-fluid (VOF) fluxes, semi-discrete energy conservation, and no small-cell restriction of the time step. The answer has three parts. (i) These requirements are *incompatible* for the Gram metric as assembled: $\mathcal{M}$ applied to a constant field gives the Gram tensors of the cells, whose sum differs from the VOF volume by $O(h^2)$ and changes as the interface moves; exact momentum and exact free stream together force that sum to be invariant (proposition 2), and a flux form with the VOF fluxes alone cannot preserve constants at all (proposition 3). (ii) Sliding the pressure points of the cut cells along their symmetry lines by $O(h^3)$ ("mass matching") makes the Gram mass equal to the VOF mass and removes the obstruction. (iii) Within the remaining freedom, every design that adds the change of the metric $\mathsf{E} = \mathcal{M} - \mathsf{W}$ explicitly to the momentum (flux form, skew form, skew redistribution) divides metric jumps by the inertia of slaved values and is unbounded near small cut cells; the bounded design is the *increment form* $\mathcal{M}^{+}\delta = \mathsf{W}^{+}\Delta u_{\mathsf{W}}$, which maps the momentum increment of the existing W transport through $\mathcal{M}^{-1}$ like a force. It preserves constants exactly in every sweep, needs one solve with the band factorisation that the projection computes anyway, and its increment is bounded by $\lambda_{\max}(\mathsf{W}, \mathcal{M}) \leq 2.7$ [M] independently of the cut volumes; a mass-proportional momentum restoration (DP) makes its $\mathcal{M}$-momentum exact with mass matching. Exact semi-discrete energy is not reachable with a bounded design for the present, piecewise-in-time, metric. **Open items**: R3 (energy) with a bounded design; gas values on faces of aperture below 0.05 at density ratio 1000 stay unbounded with DP (as with the current transport); DP's momentum restoration is verified for constant fields only; the KH, RT, capillary-wave, combined-case and cost measurements were not run (study parked by the owner). [M] marks a measurement, [I] an inference. Prototype: worktree `wt-mm-momentum`, branch `study/mm-momentum-transport`; runs and report in `mm-runs/mm_momentum/`.

Contents

1 Setting and requirements **2**

2 What the Gram metric does to a constant field **2**

3 Obstructions **3**

4 Designs **4**

5 Results **6**

5.1 Free stream 6

5.2 Small cut volumes and the time step 7

5.3 Static and translating drop 7

5.4 Cost 8

5.5 Not measured (stopped by the owner) 8
5.6 Answers 8

1 Setting and requirements

Unknowns. The phase values $u \in \mathbb{R}^N$ are the degrees of freedom of the assembly of the projection: one value per face and phase with positive staggered phase volume, one identified value on a grid-line interface face. At a time level the geometry (PLIC reconstruction, shared apertures, pressure points) gives

- the Gram metric $\mathcal{M}$: per cut cell and phase the block $\rho^\pi h^2(\mathsf{R}\mathsf{G}^+\mathsf{R}^\top + \mathsf{S})$ on the wetted sides, $\frac{1}{2}\rho^\pi h^2$ per side of an uncut cell, ties of slaved and small-aperture values to donors, lumped masses where no block gives inertia (`_assemble_cut_cell_metric!`);
- the lumped metric $\mathsf{W} = \mathrm{diag}(m)$ with the staggered phase masses $m_{(f,\pi)} = \rho^\pi |V_f|\, \bar\alpha_f^\pi$, $\bar\alpha^\pi$ the interpolated VOF fractions; the code calls them `dof_mass`;
- the excess $\mathsf{E} = \mathcal{M} - \mathsf{W}$, which vanishes outside the band of cut cells (an uncut face between two pure cells of one phase has $\mathcal{M}_{ff} = \rho h^2 = m_f$);
- the constant fields $\mathsf{Z} = [\mathsf{Z}_x^\ell, \mathsf{Z}_y^\ell, \mathsf{Z}_x^g, \mathsf{Z}_y^g]$, $(\mathsf{Z}_k^\pi)_{(f,\pi')} = 1$ if f is normal to e_k and $\pi' \in \{\pi, \text{identified}\}$, else 0.

The $\mathcal{M}$-momentum is $P = \mathsf{Z}^\top \mathcal{M} u$ (per phase and component; the mixture momentum sums the phases), the $\mathcal{M}$-energy $\frac{1}{2} u^\top \mathcal{M} u$. The projection conserves the mixture P up to the wall force and does not increase the $\mathcal{M}$-energy.

The current transport (techref ch. 6, §6.8) advances each phase value in two directional Lax–Wendroff sweeps with the dual phase volume fluxes interpolated from the VOF fluxes, in fluctuation form with the predicted staggered fractions. For this note it is the map $u \mapsto u_\mathsf{W}$ with the properties used below [I] (techref eqs. momentum:conservative, momentum:mass-identity, momentum:skew):

$$\mathsf{W}^+ u_\mathsf{W} = \mathsf{W} u - \Delta t\, \mathsf{C}_\mathsf{W}(u), \qquad \mathsf{Z}^\top \mathsf{C}_\mathsf{W}(u) = 0, \qquad \Delta t\, \mathsf{C}_\mathsf{W}(\mathsf{Z}) = -(\mathsf{W}^+ - \mathsf{W})\mathsf{Z}, \tag{1}$$

the last two per sweep (conservation and the mass identity), and for the all-centred states in the semi-discrete limit $\mathsf{C}_\mathsf{W} + \mathsf{C}_\mathsf{W}^\top = -\dot{\mathsf{W}}$. The CTU switch, the drain limiter and the fraction floor modify the states near small fractions; constants are preserved by all of them (fluctuation form).

Requirements (the owner's, in the order of the assignment).

R1 $\mathcal{M}$-momentum: $\mathsf{Z}^\top \mathcal{M}^+ u^+ = \mathsf{Z}^\top \mathcal{M} u$ on periodic or no-flux domains.

R2 Free stream with the VOF fluxes: the momentum fluxes are built from the VOF volume fluxes (mapped to the dual control volumes as now); a constant field stays constant to round-off through every sweep; the momentum of a constant field is $\rho^\pi \times$ VOF volume $\times\, u$.

R3 Energy: $\frac{\mathrm{d}}{\mathrm{d}t} \frac{1}{2} u^\top \mathcal{M} u = 0$ for the centred states, so that Lax–Wendroff dissipates $O(\Delta t)$ only.

R4 No small-cell restriction: the full-cell CFL bound of the current transport suffices for any cut volume; bounded phase values near tiny cut volumes.

2 What the Gram metric does to a constant field

Proposition 1 (Constant fields)**.** *Let every cut cell have a consistent block, $m\mathsf{N} = \mathsf{R}$ (Prop. 2 of the note `metric_options`), and let no value be lumped. Then*

(a) $(\mathcal{M}\mathsf{Z}_k^\pi)_{(f,\pi)} = \rho^\pi a_f^\pi h\,(x^*_{c_+(f)} - x^*_{c_-(f)})_k$*, for faces of both orientations ("targets" R; on a face normal to e_t, $t \neq k$, the entry is the tangential offset of the two pressure points);*

(b) $\mathsf{Z}^{\pi\top}\mathcal{M}\mathsf{Z}^{\pi} = \rho^{\pi}\sum_c \mathsf{G}_c^{\pi}$ *(2 × 2 per phase), with* $\mathsf{G}_c^{\pi} = h^2\mathsf{I}$ *or* 0 *in an uncut cell and the Gram tensor at the pressure point in a cut cell, if no value is identified (an identified grid-line value adds the diagonal entry of the other phase to both columns);*

(c) $\mathsf{Z}^{\pi\top}\mathsf{W}\mathsf{Z}^{\pi} = \rho^{\pi}V^{\pi}\,\mathsf{I}$ *on a uniform periodic mesh (*$|V_f| = h^2$*, face fractions the means of the two cell fractions),* V^{π} *the VOF volume of the phase.*

Proof. (a) is `metric_options` Prop. 3; ties vanish on constants because a value and its donor are parallel faces (the face ring). (b) Sum (a) against Z^{π}: the side terms of a cell sum to its Gram tensor (`metric_options` Prop. 4). (c) The interpolated fractions of the faces normal to e_k sum to the cell fractions, each cell contributing half of its volume to each of its two faces. □

So the $\mathcal{M}$-mass of a phase is $\rho^{\pi}\sum_c \mathsf{G}_c^{\pi}$, not $\rho^{\pi}V^{\pi}$. For a closed aperture polygon with the pressure point at the chord midpoint $\mathsf{G}_c^{\ell} = |P_c|\,\mathsf{I}$, the polygon area; with apertures averaged from two traces $|P_c| \neq \alpha_c h^2$, and the pressure point is moved within its admissible interval. Per value the two metrics differ by $O(1)$: a slaved value (aperture 0) has $(\mathcal{M}\mathsf{Z})_f = 0$ and $(\mathsf{W}\mathsf{Z})_f = m_f$, and the cross entries of (a) have no counterpart in W. [M] (`geometry.jl`, drop $R = 0.2$, $\rho = (1, 10^{-3})$, sliding through one cell):

n	mean $\frac{(\sum_c \mathsf{G}_c^{\ell})_{xx} - V^{\ell}}{V^{\ell}}$	range (slide)	$\frac{\lvert(\sum_c \mathsf{G}_c^{\ell})_{xy}\rvert}{V^{\ell}}$	mass-matched	max $\lvert s_c\rvert$
16	-3.5×10^{-3}	6.4×10^{-3}	7.1×10^{-4}	6.6×10^{-16}	$3.1 \times 10^{-2}h$
32	-4.2×10^{-4}	6.9×10^{-4}	3.6×10^{-5}	8.8×10^{-16}	$7.7 \times 10^{-3}h$
64	-1.1×10^{-4}	8.3×10^{-5}	3.3×10^{-6}	6.6×10^{-16}	$2.5 \times 10^{-3}h$

The mismatch is $O(h^2)$ relative and moves by $O(h^3)$ as the drop slides.

Mass matching. Moving x^* along the symmetry line by $s\hat{\eta}$ changes G^{ℓ} by $-s|\eta|\hat{\eta}\hat{\eta}^{\top}$ and G^{g} by the opposite (`metric_options` Prop. 2(c)). The slides of least $\sum_c s_c^2$ with $\sum_c \mathsf{G}_c^{\ell}(s_c) = \sum_c \alpha_c h^2\,\mathsf{I}$ are

$$s_c = |\eta_c|\,\langle \hat{\eta}_c\hat{\eta}_c^{\top}, \Lambda\rangle, \qquad \sum_c |\eta_c|^2\langle \hat{\eta}_c\hat{\eta}_c^{\top}, \Lambda\rangle\,\hat{\eta}_c\hat{\eta}_c^{\top} = \sum_c \big(\mathsf{G}_c^{\ell}(0) - \alpha_c h^2\mathsf{I}\big), \tag{2}$$

a 3×3 symmetric system for Λ. Its rank is the number of distinct normal directions $\pm\hat{\eta}_c$ (at most three): with two or fewer (a straight interface, or normals only along the axes) only the part of the right side in $\operatorname{span}\{\hat{\eta}_c\hat{\eta}_c^{\top}\}$ can be matched, and the remainder stays (it vanishes for a straight interface with exact apertures, not in general for averaged traces). When it is solvable, $\mathsf{Z}^{\top}\mathcal{M}\mathsf{Z} = \mathsf{Z}^{\top}\mathsf{W}\mathsf{Z}$ exactly (both phases, since $\mathsf{G}^{\ell} + \mathsf{G}^{g} = h^2\mathsf{I}$ and $\alpha^{\ell} + \alpha^{g} = 1$). The slides are $O(h^3)$ [M] (table above, "max slide"). This is the statement R2 asks for, in total: *with mass matching,* $\mathcal{M}$ *applied to a constant field carries* ρ^{π} *times the VOF volume of the phase*; per value it cannot hold (proposition 3). It costs one global 3×3 solve per assembly (serial sums: bitwise reproducible) and touches the projection through the pressure points (accuracy in section 5).

3 Obstructions

Consider any update of the form

$$\mathcal{M}^{+}u^{+} = \mathcal{M}u - L(u), \qquad L \text{ linear}. \tag{3}$$

Proposition 2 (Momentum and free stream need an invariant Gram mass)**.** (3) *conserves* $\mathsf{Z}^{\top}\mathcal{M}u$ *for every* u *iff* $\mathsf{Z}^{\top}L = 0$*, and preserves every constant field* $u = \mathsf{Z}U$ *iff* $L\mathsf{Z} = -(\mathcal{M}^{+} - \mathcal{M})\mathsf{Z}$*. Both together imply* $\mathsf{Z}^{\top}(\mathcal{M}^{+} - \mathcal{M})\mathsf{Z} = 0$*, i.e.* $\sum_c \mathsf{G}_c^{\pi}$ *is the same at both time levels. With the VOF mass conserved this is* $\mathsf{Z}^{\top}(\mathsf{E}^{+} - \mathsf{E})\mathsf{Z} = 0$*.*

Proof. The first two statements are (3) multiplied by $\mathsf{Z}^{\top}$ and evaluated at $\mathsf{Z}U$. Then $0 = \mathsf{Z}^{\top}L\mathsf{Z} = -\mathsf{Z}^{\top}(\mathcal{M}^{+} - \mathcal{M})\mathsf{Z}$; proposition 1(b),(c). □

By section 2 the obstruction is nonzero for the Gram metric as assembled $[\mathsf{M}]$: no transport can satisfy R1 and R2 exactly. With mass matching it vanishes. The condition is sharp: if $\mathsf{Z}^\top(\mathcal{M}^+ - \mathcal{M})\mathsf{Z} = 0$, the dense $L = DP^\top$ with $D = -\Delta\mathcal{M}\mathsf{Z}$, $P = \mathsf{Z}(\mathsf{Z}^\top\mathsf{Z})^{-1}$ satisfies both (locality is the issue, proposition 5).

Proposition 3 (A flux form with the VOF fluxes cannot preserve constants)**.** *Let $L = \Delta t\,\mathsf{C_W}$, the momentum fluxes of the VOF volume fluxes (1). Then (3) preserves constants iff $(\mathcal{M}^+ - \mathsf{W}^+)\mathsf{Z} = (\mathcal{M} - \mathsf{W})\mathsf{Z}$ per value, i.e. EZ is invariant. A value that enters or leaves the band (where $\mathsf{EZ} = 0$) must then have $(\mathsf{EZ})_f = 0$ already; for a moving interface every band value does so in turn, which requires $\mathcal{M}\mathsf{Z} = \mathsf{WZ}$ on the band and fails by $O(1)$ per value (section 2). (For a static geometry the condition is trivially met.)*

Proof. $\mathcal{M}^+\mathsf{Z}U = \mathcal{M}\mathsf{Z}U - \Delta t\mathsf{C_W}(\mathsf{Z}U) = \mathcal{M}\mathsf{Z}U + (\mathsf{W}^+ - \mathsf{W})\mathsf{Z}U$ by (1). □

Hence any $\mathcal{M}$-consistent design with exact free stream must add terms built from the change of the metric, $\mathsf{E}^+ - \mathsf{E}$, to the VOF fluxes; and the fluxes of R2 transport W-mass, so the question is only how the band term E enters.

4 Designs

All designs keep the existing W sweeps (VOF fluxes, Lax–Wendroff, CTU switch, drain limiter, floor) unchanged and add one solve with $\mathcal{M}^+$ at the end of the step; $\bar{u}$ denotes the extended start values on the new degrees of freedom (the frozen sources of the sweeps), u_W the values after the sweeps, and $c = \mathsf{E}u$ the old excess covector mapped to the new values (an identified value split by its mass weight; values that disappear are "orphans").

D0 current: $u^+ = u_\mathsf{W}$.

Dθ $\mathcal{M}^+u^+ = \mathcal{M}^+\bar{u} + \mathsf{W}^+(u_\mathsf{W} - \bar{u}) + (1-\theta)\,(c - \mathsf{E}^+\bar{u})$; **DI** $\theta = 1$ (increment form), **DS** $\theta = \frac{1}{2}$ (skew form), **DF** $\theta = 0$ (flux form: $\mathcal{M}^+u^+ = \mathcal{M}u - \Delta t\mathsf{C_W}(u)$).

DG DS plus a skew band coupling Y, $\mathsf{YZ} = \frac{1}{2}(\mathsf{E}^+\mathsf{Z} - (\mathsf{EZ})_{\text{mapped}})$ (least squares, minimal norm, on the graph of same-phase pairs of a cell, of the metric and of the face ring), plus the remainder $\Psi(\bar{u}) = \sum_k r_k\langle\bar{u}\rangle_k$ that keeps constants exact ($r = \frac{1}{2}\Delta\mathsf{EZ} - \mathsf{YZ}$, $\langle\cdot\rangle_k$ the mass mean of column k over the ring).

DP DI plus a mass-proportional momentum restoration: the $\mathcal{M}$-momentum change of DI less its constant part, $d_k = \mathsf{Z}_k^\top(\mathsf{E}^+\bar{u} - \mathsf{E}u) - \sum_j \mathsf{Z}_k^\top\Delta\mathsf{EZ}_j\langle\bar{u}\rangle_j$, is removed by the force $-\mathsf{WZ}_k\,d_k/\mathsf{Z}_k^\top\mathsf{WZ}_k$ (identified values excluded).

Each needs the excess of the old level (one product with the cached $\mathcal{M}$ before the sweeps) and one solve with the band factorisation of $\mathcal{M}^+$ (`_mimetic_metric_solve!`), which the viscous step and the projection of the same step reuse; DG adds an LSQR solve on the band graph.

Proposition 4 (The θ family)**.** *Let (1) hold and let no value disappear. Then for Dθ*

(a) constants: $\mathcal{M}^+(u^+ - \mathsf{Z}U) = -(1-\theta)(\mathsf{E}^+ - \mathsf{E})\mathsf{Z}U$: exact iff $\theta = 1$;

(b) momentum: $\mathsf{Z}^\top\mathcal{M}^+u^+ - \mathsf{Z}^\top\mathcal{M}u = \theta\,\mathsf{Z}^\top(\mathsf{E}^+ - \mathsf{E})u$: exact iff $\theta = 0$;

(c) energy, semi-discrete with centred states: $\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}u^\top\mathcal{M}u = (\theta - \frac{1}{2})\,u^\top\dot{\mathsf{E}}u$: exact iff $\theta = \frac{1}{2}$.

The three properties are governed by three different defects, $\Delta\mathsf{EZ}$ (a), $\mathsf{Z}^\top\Delta\mathsf{E}u$ (b) and $u^\top\dot{\mathsf{E}}u$ (c); when all three are nonzero (generic in the band, measured) no θ satisfies two of R1–R3. When the degrees of freedom change, (b) reads exactly $\theta(\mathsf{Z}^{+\top}\mathsf{E}^+\bar{u} - \mathsf{Z}^\top\mathsf{E}u) - (1-\theta)\,\mathrm{orph}$ with the excess of the values that disappear.

Proof. Write the update as $\mathcal{M}^+u^+ - \mathcal{M}u = \mathsf{W}^+u_\mathsf{W} - \mathsf{W}u + \theta(\mathsf{E}^+ - \mathsf{E})u$ (expand $\mathcal{M}^+\bar{u} = \mathsf{W}^+\bar{u} + \mathsf{E}^+\bar{u}$). (a) For $u = \mathsf{Z}U$, $u_\mathsf{W} = \mathsf{Z}U$, so the right side is $(\mathsf{W}^+ - \mathsf{W})\mathsf{Z}U + \theta\Delta\mathsf{E}\mathsf{Z}U = \Delta\mathcal{M}\mathsf{Z}U - (1-\theta)\Delta\mathsf{E}\mathsf{Z}U$. (b) Multiply by $\mathsf{Z}^\top$ and use $\mathsf{Z}^\top\mathsf{W}^+u_\mathsf{W} = \mathsf{Z}^\top\mathsf{W}u$. (c) In the limit, $\frac{\mathrm{d}}{\mathrm{d}t}(\mathcal{M}u) = -\mathsf{C}_\mathsf{W}u + \theta\dot{\mathsf{E}}u$, so $\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}u^\top\mathcal{M}u = u^\top\frac{\mathrm{d}}{\mathrm{d}t}(\mathcal{M}u) - \frac{1}{2}u^\top\dot{\mathcal{M}}u = -u^\top\mathsf{C}_\mathsf{W}u + \theta u^\top\dot{\mathsf{E}}u - \frac{1}{2}u^\top\dot{\mathsf{W}}u - \frac{1}{2}u^\top\dot{\mathsf{E}}u$ and $u^\top\mathsf{C}_\mathsf{W}u = -\frac{1}{2}u^\top\dot{\mathsf{W}}u$. □

[M] (`smoke.jl`): one step of each design on a drop, $\rho = (1, 10^{-3})$, $n = 32$, reproduces (b) to 2×10^{-17} absolute (momentum ≈ 0.13), orphans included.

Proposition 5 (Skew redistribution)**.** *If $\mathsf{Z}^\top(\mathsf{E}^+ - \mathsf{E})\mathsf{Z} = 0$, there is a skew Y with $\mathsf{Y}\mathsf{Z} = \frac{1}{2}(\mathsf{E}^+ - \mathsf{E})\mathsf{Z}$, and DG with $\Psi = 0$ satisfies (a), (b) and (c) of proposition 4 exactly. On a graph whose edges join the values of a phase in the cut cells, the kernel of the edge-to-value map $\mathsf{Y} \mapsto \mathsf{Y}\mathsf{Z}$ (transposed) contains the symmetric 2×2 fields $\{\mathsf{Z}\mathsf{S}\}$ per connected component (it equals them only if the x- and the y-values of a component are each connected through same-axis edges), so a local Y needs the condition per component: one global mass match does not suffice when the band has several components (checked counterexample in `proof_review/mimetic-momentum-transport`).*

Proof. Existence: with $D = \frac{1}{2}\Delta\mathsf{E}\mathsf{Z}$ and $P = \mathsf{Z}(\mathsf{Z}^\top\mathsf{Z})^{-1}$, $\mathsf{Y} = DP^\top - PD^\top$ is skew and $\mathsf{Y}\mathsf{Z} = D - P(\mathsf{Z}^\top D)^\top = D$ because $\mathsf{Z}^\top D = 0$. (a) $-\frac{1}{2}\Delta\mathsf{E}\mathsf{Z} + \mathsf{Y}\mathsf{Z} = 0$. (b) $\mathsf{Z}^\top(\frac{1}{2}\Delta\mathsf{E}u + \mathsf{Y}u) = \frac{1}{2}(\Delta\mathsf{E}\mathsf{Z})^\top u - (\mathsf{Y}\mathsf{Z})^\top u = 0$ ($\Delta\mathsf{E}$ symmetric, Y skew). (c) $u^\top\mathsf{Y}u = 0$ and the $\dot{\mathsf{E}}$ terms of proposition 4(c) cancel at $\theta = \frac{1}{2}$. Kernel: an edge (f, g) maps y to $+y$ in row $(f, \mathrm{axis}(g))$ and $-y$ in row $(g, \mathrm{axis}(f))$; $r_{(f,k)} = \mathsf{S}_{\mathrm{axis}(f),k}$ with S symmetric is annihilated. □

Proposition 6 (Increment form: bounded, no small-cell restriction)**.** *The DI increment $\delta = u^+ - \bar{u} = (\mathcal{M}^+)^{-1}\mathsf{W}^+(u_\mathsf{W} - \bar{u})$ satisfies*

$$\|\delta\|^2_{\mathcal{M}^+} \le \lambda_{\max}(\mathsf{W}^+, \mathcal{M}^+)\,\|u_\mathsf{W} - \bar{u}\|^2_{\mathsf{W}^+}, \qquad \|\delta\|_\infty \le \|(\mathcal{M}^+)^{-1}\mathsf{W}^+\|_\infty\,\|u_\mathsf{W} - \bar{u}\|_\infty,$$

and preserves constants exactly in every sweep. The time step enters only through the W sweeps, which are bounded under the full-cell CFL of the current transport.

Proof. $\|\delta\|^2_{\mathcal{M}} = v^\top\mathsf{W}\mathcal{M}^{-1}\mathsf{W}v \le \lambda_{\max}(\mathcal{M}^{-1/2}\mathsf{W}\mathcal{M}^{-1/2})\,v^\top\mathsf{W}v$ with $v = u_\mathsf{W} - \bar{u}$. Constants: $u_\mathsf{W} = \bar{u} = \mathsf{Z}U$. □

[M] over random drops and lines with cut volumes down to 10^{-12} (`geometry.jl`):

geometry	n	floor	min $\min(\alpha, 1-\alpha)$	$\lambda_{\max}(\mathsf{W}, \mathcal{M})$	$\|\mathcal{M}^{-1}\mathsf{W}\|_\infty$	$\min(\mathcal{M}^{-1}\mathsf{W})$	$\max_f \mu_f$
drop	16	$\sqrt{\epsilon}$	9.5×10^{-8}	2.63	4.98	−0.80	1.00
drop	32	$\sqrt{\epsilon}$	5.3×10^{-8}	2.54	4.97	−0.81	1.00
drop	16	10^{-12}	3.0×10^{-12}	2.63	4.95	−0.77	1.00
drop	32	10^{-12}	4.3×10^{-12}	2.41	4.53	−0.68	1.00
lines	16	$\sqrt{\epsilon}$	1.0×10^{-6}	2.22	4.28	−0.25	1.00
lines	32	$\sqrt{\epsilon}$	1.7×10^{-8}	2.18	4.00	−0.25	1.00
lines	16	10^{-12}	1.3×10^{-12}	2.22	4.28	−0.25	1.00
lines	32	10^{-12}	1.0×10^{-12}	2.06	3.77	−0.25	1.00

40 random configurations per row (radius, sliver depth 10^{-1}–$10^{-9}h$, line direction), worst value over the row, $\rho = (1, 10^{-3})$; $\mu_f = |((\mathcal{M} - \mathsf{W})\mathsf{Z})_f| / \max(m_f, \mathcal{M}_{ff})$ is the per-value mismatch of the constant fields. The algebra alone gives no cut-volume independence ($\lambda_{\max} \ge \max_f m_f/\mathcal{M}_{ff}$); it comes from the assembly: the tie of a slaved value carries its own staggered mass, so the row of $\mathcal{M}$ scales with the row of W (measured, no analytic bound). The entries of $\mathcal{M}^{-1}\mathsf{W}$ are not all non-negative (min ≈ -0.8): the map is bounded but not monotone, and DI can create small new extrema near the interface.

Remark 7 (Why the designs with E terms are unbounded)**.** *DF, DS and DG add* $(\mathcal{M}^+)^{-1}(c - \mathsf{E}^+\bar{u})$ *(and* $\mathsf{Y}\bar{u}$*). The row of a slaved value (aperture* 0*) of* $\mathcal{M}^+$ *is its tie, of weight equal to its staggered mass, as small as* 10^{-14} *[M]; the same row of* $c - \mathsf{E}^+\bar{u}$ *carries the excess of the old level, which belonged to a value with a Gram block or another donor.* E *changes by* $O(1)$*, not* $O(\Delta t)$*, when a value changes its status (wetted* $\leftrightarrow$ *slaved, donor switch, lumping, grid-line identification, cut* $\leftrightarrow$ *uncut), so the quotient is not small for small* Δt*. For constants these terms cancel exactly (proposition 4(a), proposition 5); for any other field they do not. [M]: one to five steps with a vortex perturbation gave corrections* $|u^+ - u_{\mathsf{W}}|$ *of* 64 *(DS),* 1.2×10^3 *(DF),* 4×10^3*–*6×10^4 *(DG), all on slaved values, against* 0.13 *for DI (*`smoke.jl`*).*

Proposition 8 (Momentum of the increment form)**.** *For DI,* $P^N - P^0 = \mathsf{Z}^\top\mathsf{E}^N u^{N-1} - \mathsf{Z}^\top\mathsf{E}^0 u^0 - \sum_{n=1}^{N-1} \mathsf{Z}^\top\mathsf{E}^n(u^n - u^{n-1})$ *(plus the* W*-momentum defects of the sweeps): the drift is bounded by* $\|\mathsf{E}\|$ *times the variation of* u *in time and does not grow for a steady field. With mass matching the constant part vanishes, and DP removes the rest:* $\mathsf{Z}^\top\mathcal{M}^+u^+ = \mathsf{Z}^\top\mathcal{M}u$ *exactly, constants exact.*

Proof. Sum proposition 4(b) at $\theta = 1$ by parts. DP: the restoration force has $\mathsf{Z}_k^\top$-moment $-d_k$, and d_k equals the DI change less $\sum_j \mathsf{Z}_k^\top \Delta\mathsf{E}\mathsf{Z}_j\langle\bar{u}\rangle_j$, which is zero with mass matching; for a constant field $d_k = 0$ (proposition 4(a)). The force is mass-proportional, so proposition 6 bounds its effect by $\|\mathcal{M}^{-1}\mathsf{W}\|_\infty |d_k|/\mathsf{Z}_k^\top\mathsf{W}\mathsf{Z}_k$, a uniform velocity of the size of the momentum defect over the phase mass. Exactly, DP changes the momentum by $\sum_j \mathsf{Z}^\top\Delta\mathsf{E}\mathsf{Z}_j\langle\bar{u}\rangle_j$, provided the sweeps conserve the W-momentum (no clamped or snapped fractions). □

Derivation check. An independent check of the propositions of sections 2 to 4 and of (2) (randomised NumPy checks, 653 passing) is archived with the proof reviews of the audit, folder `mimetic-momentum-transport`; its corrections (identified values in proposition 1(b), the rank of (2), the wording of proposition 3 and of the trilemma, the per-component condition for a local Y, the measured, not analytic, cut-volume independence of the increment bound) are incorporated.

5 Results

All runs: uniform periodic meshes, `MimeticProjection`, the default Lax–Wendroff transport with the CTU switch, drivers in `mm-runs/mm_momentum/` (`freestream.jl`, `smallcell.jl`, `drop.jl`, `geometry.jl`, `smoke.jl`; outputs `*.txt`). CFL is the fraction of the full-cell VOF bound of `stable_timestep` (the WY bound admits $0 < \mathrm{CFL} \le \frac{1}{2}$).

5.1 Free stream

[M] Uniform $U \in \{(1, 0.37), (0, 1)\}$ in both phases, a drop ($R = 0.2$, cap 10^{-6} of a cell entering a new row) or a band between the lines $x + 2y = c \pm 0.17$ (one line $10^{-6}h$ from a grid vertex), $n = 32$, 30 steps, $\rho_g \in \{1, 10^{-3}\}$, CFL 0.1 and 0.45, cell fraction floor $\sqrt{\epsilon}$ and 10^{-12}; maximum over the runs:

case	design	$\frac{\max\lvert u-U\rvert}{\lvert U\rvert}$, transport	after the step	$\frac{\max\lvert\Delta P_{\mathrm{mix}}\rvert}{\rho V\lvert U\rvert}$	min $\min(\alpha, 1-\alpha)$
drop, $\sqrt{\epsilon}$	D0, DI	0	0	2.1×10^{-4}	1.0×10^{-7}
	DP+mm	1.4×10^{-16}	1.4×10^{-16}	3.0×10^{-12}	
	DG+mm	5.5×10^{-5}	5.5×10^{-5}	1.0×10^{-11}	
	DS / DF	4.6×10^3 / 8.6×10^4		1.1 / 2.2×10^{-3}	
drop, 10^{-12}	D0, DI	0	0	2.1×10^{-4}	5.3×10^{-10}
	DP+mm	1.4×10^{-16}	1.4×10^{-16}	2.4×10^{-14}	
	DG+mm	310	310	2.4×10^{-5}	
	DS / DF	1.6×10^7 / 1.2×10^5		1.9 / 2.2×10^{-3}	
band, $\sqrt{\epsilon}$	D0, DI	0	0	3.7×10^{-8}	1.3×10^{-7}
	DP+mm	2.6×10^{-16}	2.6×10^{-16}	2.4×10^{-8}	
	DG+mm	1.8×10^{-3}	1.8×10^{-3}	2.4×10^{-8}	
	DS / DF	6.0×10^3 / 4.7×10^5		0.23 / 1.4×10^{-2}	

D0 and DI keep constants bitwise (the W sweeps are in fluctuation form and the DI right side is exactly zero), through every sweep and every step, at all CFL numbers and cut volumes tried; DP to round-off. Their $\mathcal{M}$-momentum of the constant field drifts by the obstruction of proposition 2 (2×10^{-4} in 30 steps for the drop) unless the pressure points are mass-matched, after which it is round-off; the band case keeps 2.4×10^{-8}, [I] the VOF's own volume change by fraction snapping (an O(floor) effect, the same for every design). DG preserves constants only up to round-off amplified by the inverse inertia of slaved values (the cancellation in proposition 5 is exact in arithmetic, not in floating point); DS and DF do not preserve constants (propositions 3 and 4). The CFL 0.9 runs of `freestream.txt` are rejected by the WY bound check, not by the designs.

5.2 Small cut volumes and the time step

[M] A drop carried by $U = (1, 0.37)$ plus a solenoidal vortex of amplitude 0.3 crosses the box once ($t = 1$, $n = 32$), so the interface sweeps through near-empty faces. Steps to $t = 1$, maximum phase speed over the run relative to the initial maximum on values whose phase aperture is at least 0.05 ("wet", the floor of the static-drop test) and over all values ("all"); a run stops (†) when $E > 100E_0$ or a value exceeds $10^4 u_0$:

			D0 (current)			DI			DP + mass matching			DG+mm	DF
ρ_g	floor	CFL	steps	wet	all	steps	wet	all	steps	wet	all	stopped at step	
1	$\sqrt{\epsilon}$	0.10	538	1.01	1.01	542	1.03	2.43	541	1.02	1.14	3	126
1	$\sqrt{\epsilon}$	0.25	215	1.01	1.01	216	1.02	1.05	216	1.03	1.05	1	24
1	$\sqrt{\epsilon}$	0.50	108	1.01	1.00	108	1.02	1.04	108	1.02	1.05	21	42
1	10^{-12}	0.10	538	1.01	1.01	542	1.03	2.43	541	1.02	1.15	3	54
1	10^{-12}	0.25	215	1.01	1.01	216	1.02	1.05	216	1.03	1.05	1	43
1	10^{-12}	0.50	108	1.01	1.00	108	1.02	1.04	108	1.02	1.05	7	55
10^{-3}	$\sqrt{\epsilon}$	0.10	735	6.13	204	1228	1.69	200	1177	1.79	159	5	944
10^{-3}	$\sqrt{\epsilon}$	0.25	271	3.74	1.0×10^3	708	1.76	171	610	2.19	162	1	5
10^{-3}	$\sqrt{\epsilon}$	0.50	124	5.84	610	363	3.35	263	402	2.03	110	6	2
10^{-3}	10^{-12}	0.10	827	5.30	637	1440	1.88	370	1500	1.71	5.2×10^3	5	120
10^{-3}	10^{-12}	0.25	336	6.33	360	950	2.60	1.3×10^3	822	2.11	373	1	5
10^{-3}	10^{-12}	0.50	145†	6.13	1.0×10^4	491	1.81	1.9×10^3	622	5.31	352	6	2

$\mathcal{M}$-energy maximum over the run relative to its initial value: D0 1.0001–1.008, DI 1.0008–1.0096, DP+mm $\leq$ 1.0000 (to the four printed digits in every run).

- DI and DP run at every CFL up to the WY maximum $\frac{1}{2}$ with cut volumes down to 10^{-12}: no small-cell restriction, as proposition 6 predicts. DG and DF fail in 1–126 steps at every CFL (remark 7); smaller steps do not help, because the metric jumps do not scale with Δt.
- At $\rho_g = 1$ the values stay bounded for all three (wet $\leq$ 1.03); DI adds new extrema on small-aperture values (all $\leq$ 2.43), the non-monotone $\mathcal{M}^{-1}$W of proposition 6.
- At $\rho_g = 10^{-3}$ the current method already has gas values of 10^2–$10^4 u_0$ on faces with aperture below 0.05 (the known light-phase velocities on small apertures of the projection, `report_mmd.md`); D0 stopped on one at CFL 0.5, floor 10^{-12}. DI and DP lower the wet maxima (1.7–3.4 and 1.7–5.3 against 3.7–6.3 for D0) but do not bound the small-aperture values: DP reached $5.2 \times 10^3 u_0$ at CFL 0.1, floor 10^{-12} (**open**).
- At $\rho_g = 10^{-3}$ DI and DP take 1.6–3.4 times more steps than D0 for the same CFL fraction: the bound of `stable_timestep` is the full-cell VOF bound evaluated with the phase velocities, and larger gas values on partially wet faces lower it [I]. At $\rho_g = 1$ the step counts agree to 1%.

5.3 Static and translating drop

[M] `static_droplet_case(:mimetic, ...)` of the test, $t = 0.4$; and the same drop in a periodic box, at rest or translating with $U = (1, 0.37)$, $\max|u - U|$ on mixed faces with phase aperture above 0.05, first half → second half of the run ($n = 32$):

	D0	D0+mm	DI	DP+mm
static DR 1, La ∞: $E_{\text{late}}/E_{\text{early}}$	2.18	2.18	2.21	2.21
static DR 1000, La ∞: $E_{\text{late}}/E_{\text{early}}$	1.68	1.64	1.71	1.67
static DR 1, La 1200: gas	$4.11 \to 7.27\,(10^{-3})$	$4.11 \to 7.26$	$4.11 \to 7.26$	$4.11 \to 7.28$
static DR 1000, La 1200: gas	$1.71 \to 2.86\,(10^{-2})$	$1.70 \to 2.86$	$1.70 \to 2.88$	$1.70 \to 2.89$
periodic DR 1, $U = 0$	$7.8 \to 9.6\,(10^{-3})$	$7.8 \to 9.6$	$7.8 \to 9.6$	$7.8 \to 9.6$
periodic DR 1, $U = (1, 0.37)$	$0.191 \to 0.178$	$0.191 \to 0.178$	$0.179 \to 0.174$	$0.179 \to 0.174$
periodic DR 1000, $U = 0$	$1.7 \to 2.8\,(10^{-2})$	$1.7 \to 2.8$	$1.7 \to 2.9$	$1.7 \to 2.8$
periodic DR 1000, $U = (1, 0.37)$	$0.49 \to 1.78$	$0.49 \to 1.78$	$0.76 \to 1.32$	$0.74 \to 1.56$

No design changes the static drop materially (its growth at $n = 32$ is the capillary instability of `static_drop_stability`, sec. zero). Translation raises the spurious currents 20-fold (DR 1) and 60-fold (DR 1000) for *every* design, the current one included: the loss of Galilean invariance does not come from the metric of the transport [I] (constants are transported exactly, section 5.1); it is open.

5.4 Cost

[M] ($n = 32$, drop, `smoke.jl` and `smallcell.jl`): the band has 324–354 values; one solve with the existing band factorisation takes 6–10 μs, the snapshot and the products of the prototype (which rebuilds the sparse metric each step) 0.6 ms; the LSQR of DG 2–3.5 ms. Wall time per step with the diagnostics of the drivers: D0 8.0–9.5 ms, DI 8.6–10.3 ms, DP 8.8–11.2 ms. In production the extra cost is one back-substitution and two metric products per step (the factorisation is shared with the viscous step and the projection).

5.5 Not measured (stopped by the owner)

The KH (inviscid, momentum and energy drift, energy versus Δt), RT DR3/DR1000 $n = 32$, capillary waves DR1/DR1000 $n = 16/32/64$, combined case and the per-case timing runs were prepared (`cases.jl`, `queue.sh`) but not run: the study was parked. The energy statement of R3 is therefore only analysed (proposition 4 and remark 7), not measured as $\Delta t \to 0$.

5.6 Answers

1. *R1 and R2 together* are impossible for the Gram metric as assembled (proposition 2): its mass differs from the VOF mass by $O(h^2)$ and moves with the interface. Mass matching of the pressure points ($O(h^3)$ slides, one 3×3 solve) removes the obstruction; the static drop is unchanged by it to 3%.

2. *Flux forms* with the VOF fluxes cannot preserve constants (proposition 3): per value the Gram metric and the VOF masses differ by $O(1)$.

3. *Bounded designs.* Any design that adds the metric change $\mathsf{E}^+ - \mathsf{E}$ to the momentum (DF, DS, DG) is unbounded at slaved values and fails R4 for every Δt. The increment form DI is bounded independently of the cut volume ($\lambda_{\max}(\mathsf{W}, \mathcal{M}) \leq 2.7$), exact for constants in every sweep, and costs one back-substitution; DP adds exact $\mathcal{M}$-momentum with mass matching.

4. *R3* (exact semi-discrete energy) needs $\theta = \frac{1}{2}$ or the skew redistribution, i.e. explicit E terms: not reachable with a bounded design for the present metric, whose assembly changes discretely in time (status changes of values). **Open**: a metric that is continuous in time at the level of single values (no donor switches, continuous slaving) is the prerequisite for DS/DG.

5. *Open:* gas values on faces with aperture below 0.05 at density ratio 1000 are unbounded with D0, DI and DP alike; Galilean invariance of the capillary drop.